

Trade Study TS-004

Ground Control Station Power Distribution Architecture Selection

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

Prepared By: Serena Green

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1. Objective

The objective of this trade study is to evaluate candidate electrical power distribution architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the architecture that best satisfies the project's operational, technical, and safety requirements. The power distribution subsystem is responsible for safely delivering electrical power from the portable battery to the computer hardware, display, communications equipment, cooling system, and future mission electronics while maintaining reliable operation during field deployment.

Because the Ground Control Station is intended to function as a portable field computing platform, the selected power architecture must provide stable voltage regulation, electrical protection, maintainability, and sufficient expansion capability to support future subsystem integration. The architecture must also minimize wiring complexity while allowing individual components to be serviced or replaced without requiring significant redesign of the overall electrical system.

In addition to satisfying current operational requirements, the selected architecture should support future enhancements including additional communications hardware, AI accelerators, external sensors, and renewable charging capability while maintaining safe operation under portable field conditions.

2. Design Requirements

The Ground Control Station power distribution architecture must safely distribute electrical power to all mission-critical subsystems while maintaining stable operating voltages and protecting sensitive electronic equipment from electrical faults. Because the system will operate from a portable battery during field missions, the architecture must maximize reliability while remaining simple to troubleshoot, repair, and expand.

Electrical safety was considered a primary design objective throughout the evaluation process. The selected architecture must incorporate appropriate circuit protection, switching capability, standardized connectors, and voltage regulation while supporting future subsystem expansion without requiring redesign of the primary electrical distribution system.

Requirement	Value
Stable Output Voltage	Required
Fuse Protection	Required
Master Power Switch	Required

Portable Operation	Required
Expandable	Required
Maintainable	Required
Modular Wiring	Required
XT60 Connectors	Required
DC Boost Regulation	Required
Future Solar Compatibility	Desired
Budget	<\$150 Preferred

3. Alternatives Considered

Alternative A

Direct Battery Connection

A direct battery connection supplies power from the battery to each subsystem with minimal electrical distribution hardware. Although this approach minimizes component count and initial cost, it provides limited protection against electrical faults and complicates future subsystem expansion.

Advantages

- Lowest cost
- Minimal wiring
- Simple construction
- Few components

Disadvantages

- Limited circuit protection
- Difficult troubleshooting
- Poor scalability
- Reduced maintainability

Alternative B

Modular Fused Distribution Panel Architecture

A modular power distribution architecture routes battery power through a master switch, fuse protection, voltage regulation, and a centralized distribution panel before supplying individual subsystems. This configuration isolates electrical faults, simplifies maintenance, and supports future expansion.

Advantages

- Improved electrical safety
- Modular architecture
- Easy troubleshooting
- Expandable
- Supports future subsystem additions
- Professional wiring organization

Disadvantages

- Additional components required
- Slightly higher cost
- Increased wiring complexity

Alternative C**Commercial DC Power Hub**

Commercial DC distribution hubs provide integrated fuse protection and multiple output circuits within a compact package. These units simplify installation while reducing custom wiring.

Advantages

- Professional appearance
- Integrated protection
- Reduced fabrication effort
- Compact design

Disadvantages

- Higher acquisition cost
- Limited customization
- Fixed output configuration

Alternative D**PCB-Based Custom Distribution Board**

A custom printed circuit board distributes power through dedicated traces, connectors, and protection devices specifically designed for the Ground Control Station.

Advantages

- Highly integrated
- Compact
- Professional electrical layout
- Easily repeatable

Disadvantages

- Highest design complexity
- Requires PCB fabrication
- Difficult to modify during prototype development
- Increased development time

4. Evaluation Criteria

The candidate power distribution architectures were evaluated using a weighted decision matrix based on electrical safety, maintainability, expandability, implementation complexity, and project cost. Evaluation criteria were selected to ensure the selected architecture supports both the immediate operational requirements of the prototype and future subsystem integration activities.

Electrical safety received the highest weighting because the power distribution subsystem supplies every major Ground Control Station component and therefore represents a single point of failure. Expandability and maintainability were also assigned significant weighting because the electrical architecture is expected to evolve throughout prototype development. Cost and implementation complexity were considered to ensure the selected solution remained practical within the available project resources.

Criterion	Weight
Electrical Safety	25%
Expandability	20%
Maintainability	20%
Cost	15%
Ease of Integration	10%
Reliability	10%

5. Decision Matrix

Criteria	Weight	Direct Wiring	Fused Distribution	Commercial Hub	Custom PCB
Electrical Safety	0.25	2	5	5	5
Expandability	0.20	2	5	3	5

Maintainability	0.20	2	5	4	2
Cost	0.15	5	4	3	1
Ease of Integration	0.10	4	4	5	2
Reliability	0.10	3	5	5	4

Each alternative was evaluated using a qualitative scoring scale ranging from one to five, where a score of one represents poor performance relative to the evaluation criterion and a score of five represents excellent performance. Scores were assigned using engineering judgment, anticipated integration effort, electrical safety considerations, expected lifecycle performance, and compatibility with the operational requirements established for the TOMTOM Ground Control Station.

The weighted decision matrix provides a structured and repeatable methodology for comparing candidate electrical architectures while reducing subjective bias during the design process. By applying weighting factors to each criterion, the evaluation emphasizes characteristics most critical to safe and reliable system operation while allowing lower-priority considerations to contribute appropriately to the final engineering decision.

6. Weighted Scores

Alternative	Total Score
Direct Battery Connection	2.75
Centralized Fused Distribution Panel	4.75
Commercial DC Power Hub	4.10
Custom PCB Distribution	3.55

Weighted scores were calculated by multiplying the score assigned to each evaluation criterion by its associated weighting factor and summing the resulting values for each candidate architecture. The resulting score represents the overall suitability of each alternative relative to the established project requirements.

Although the weighted score serves as the primary decision metric, the final engineering decision also considered implementation risk, future subsystem expansion, electrical safety, hardware availability, and compatibility with the Ground Control Station architecture. Consequently, the selected architecture represents the solution providing the best overall balance between safety, reliability, maintainability, and lifecycle value rather than simply the highest numerical score.

7. Discussion

The centralized fused power distribution architecture achieved the highest weighted score among the evaluated alternatives and was selected as the preferred electrical architecture for the TOMTOM Ground Control Station. Although this approach requires additional electrical components and slightly increases wiring complexity compared to direct battery connections, it provides significantly greater safety, maintainability, and flexibility for future system expansion.

The selected architecture organizes the electrical system around a master power switch, fused distribution, voltage regulation, and standardized connectors, allowing individual subsystems to be isolated for maintenance or troubleshooting without affecting the remainder of the system. This modular approach simplifies future upgrades and reduces integration risk as additional communications equipment, sensors, AI hardware, and mission electronics are incorporated into the Ground Control Station.

The modular architecture also supports the project's verification strategy by allowing individual electrical subsystems to be energized, tested, isolated, and serviced independently. This reduces troubleshooting time during integration and simplifies future hardware modifications without requiring redesign of the complete electrical system.

Unlike custom printed circuit boards, which require significant design effort and are difficult to modify during prototype development, the selected architecture remains flexible throughout the iterative engineering process. Likewise, although commercial distribution hubs reduce wiring effort, their fixed configurations limit customization and future adaptability. The selected architecture therefore provides the best overall balance between electrical safety, system flexibility, implementation effort, and long-term maintainability while supporting continued development of the TOMTOM Ground Control Station.

8. Decision

Selected Alternative: Modular Fused Power Distribution Architecture

Rationale

The selected architecture provides the best balance between:

- Electrical safety
- Expandability
- Maintainability
- Reliability
- Cost
- Long-term flexibility

The centralized fused power distribution architecture establishes the electrical baseline for the TOMTOM Ground Control Station and provides a safe, modular, and maintainable foundation for future subsystem integration, verification testing, and continued prototype development.

The selected modular fused power distribution architecture establishes the electrical baseline configuration for the TOMTOM Ground Control Station. This architecture provides a safe, maintainable, and expandable electrical foundation that supports prototype integration, subsystem verification, and future capability growth throughout continued system development.